



URBAN SOUND CLASSIFICATION & NOISE MONITORING

2025 WINTER GLOBAL PROGRAM IN BELLEVUE COLLEGE, WASHINGTON

PRESENTED BY

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Target A - (re)Construction Co.

Pain Point - Lawsuit Risk

Direct legal costs

- Last 20 years of environmental lawsuit : **Over 20,000**
- Avg. term of lawsuit : **8 - 18 months**
- Avg. Attorney Costs : **\$300 – \$450/hour**
- Total Costs : **\$50,000 - \$100,000**

Potential compensation or settlement

- Nuisance value : **\$5,000 – \$20,000**
- Continuous or Severe noise: **Over \$100,000**
- Case :
 - **Crosstex (2016, Texas)**
 - **\$2,000,000**
 - **Bensenville (1999, Illinois)**
 - **Provided financial support** for soundproofing costs
 - **Hoboken Gym (2023, New Jersey)**
 - Operations shut down
 - Pay **\$100,000** in punitive damages

Target B - Citizen



Person 1-Meeting Room Renter

Needs a quiet time slot
But “quiet” is not guaranteed

- Books a meeting room for an important call
- Noise varies by time + location (traffic, construction, events)
- No reliable data → wasted time, poor meeting quality

Target B - Citizen

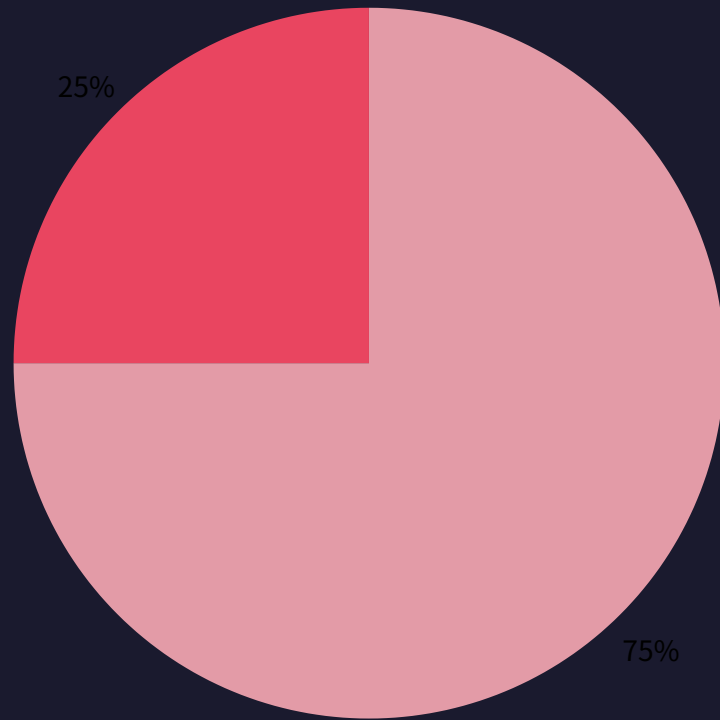


Person 2-House Seeker

Must choose a home
But “quiet” is not guaranteed

- Visits the apartment once, but noise changes at night/weekends
- Ads say “quiet neighborhood,” but it’s not measurable
- No evidence → risk of regret after signing a lease

Target B - Citizen



20–25% of urban residents experience high noise annoyance

-European Environment Agency

Urban noise arises from transportation, construction, neighborhood activities, and commercial or leisure spaces. When such noise is continuous and beyond individual control, its impact on stress is significantly amplified

Environmental noise poses a significant risk to public health

Even at the same sound level (dB), noise that individuals perceive as **uncontrollable** leads to significantly greater stress responses and adverse health effects.

-WHO

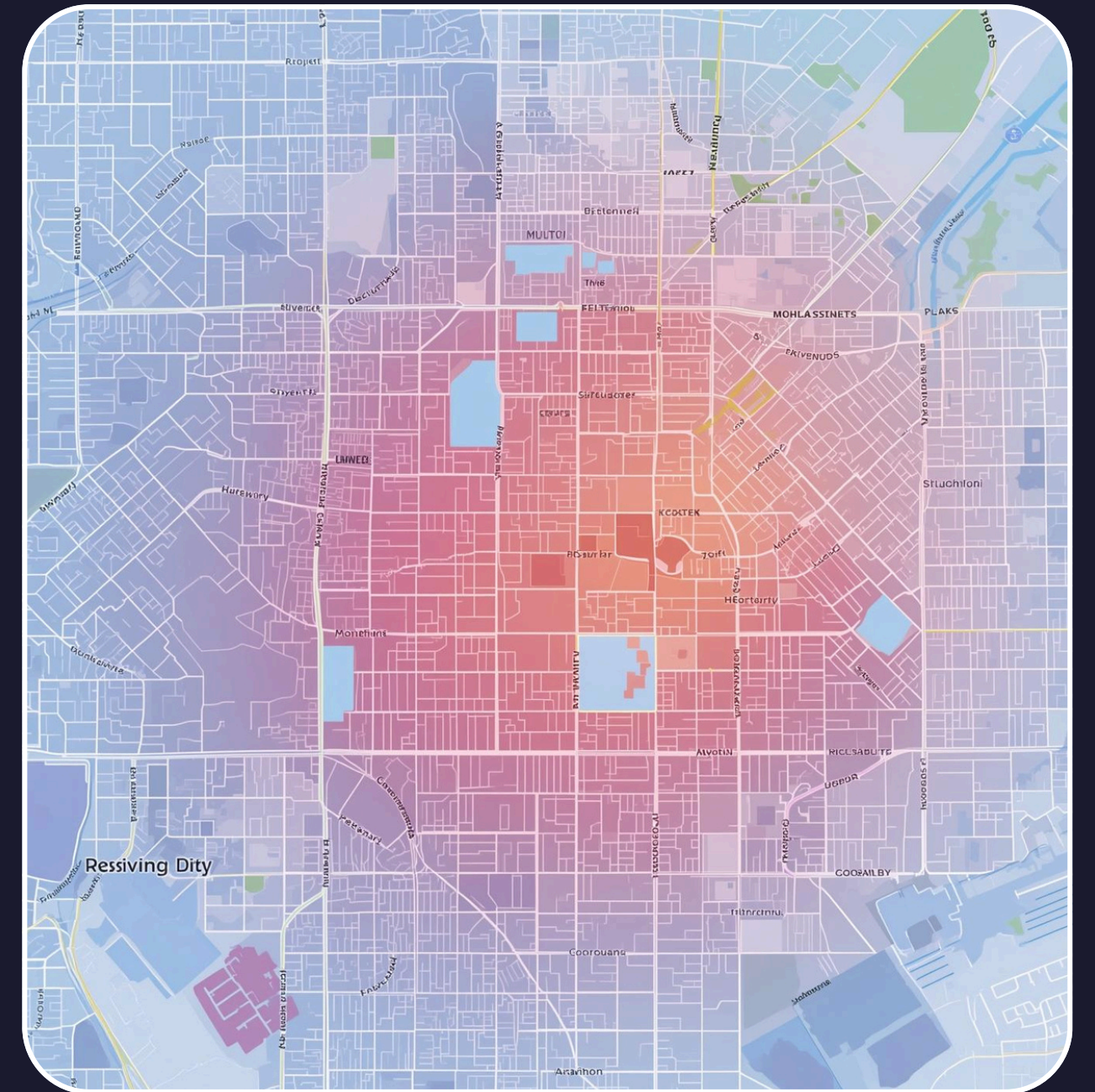


Sound Scout Solutions

Sound Scout will provide both companies and organizations with :

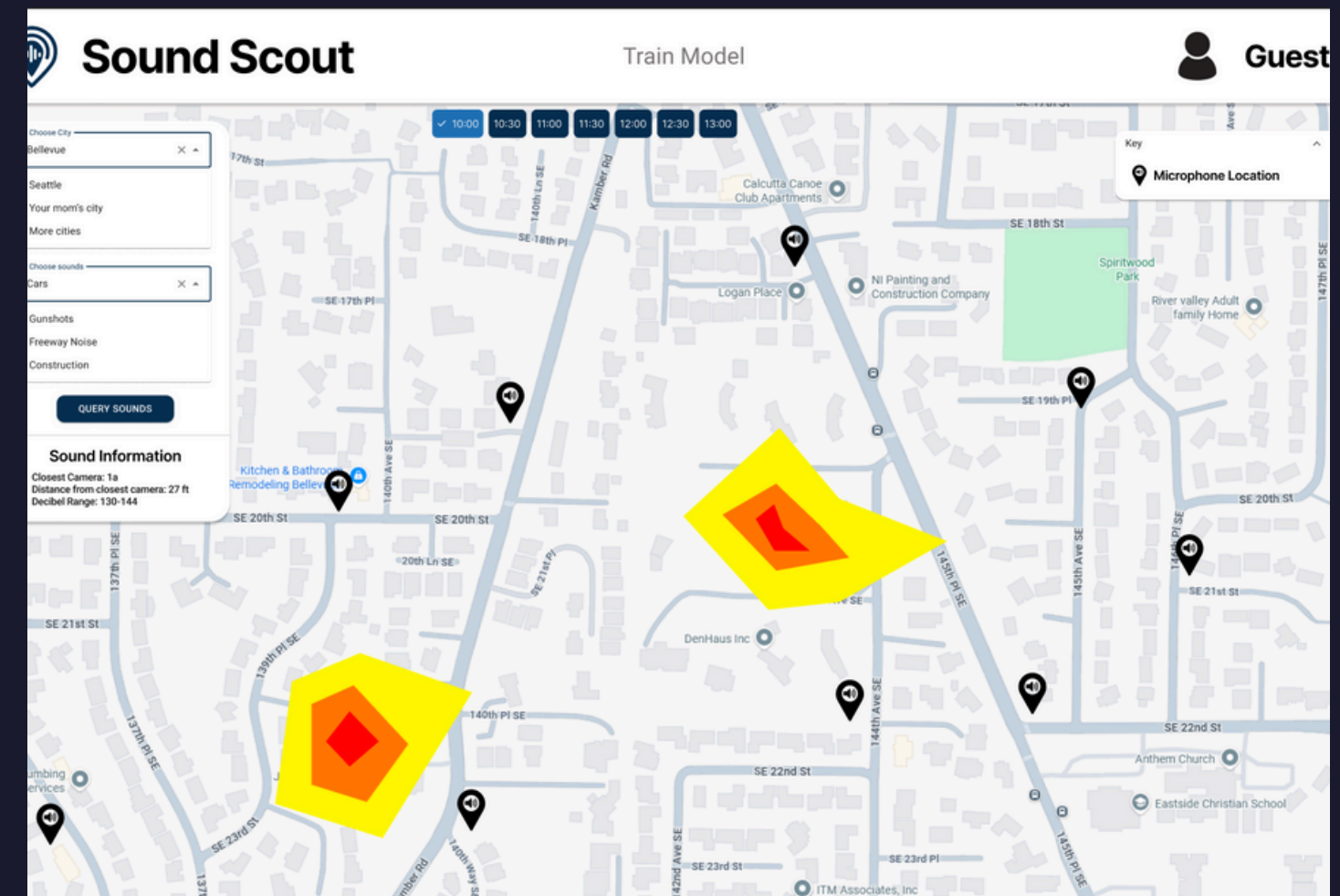
- valuable sound level
- classification information

This will be done through a Machine Learning model that will take the sound information from locally placed microphones and analyse the data and produce a heat map.



Sound Scout Backend

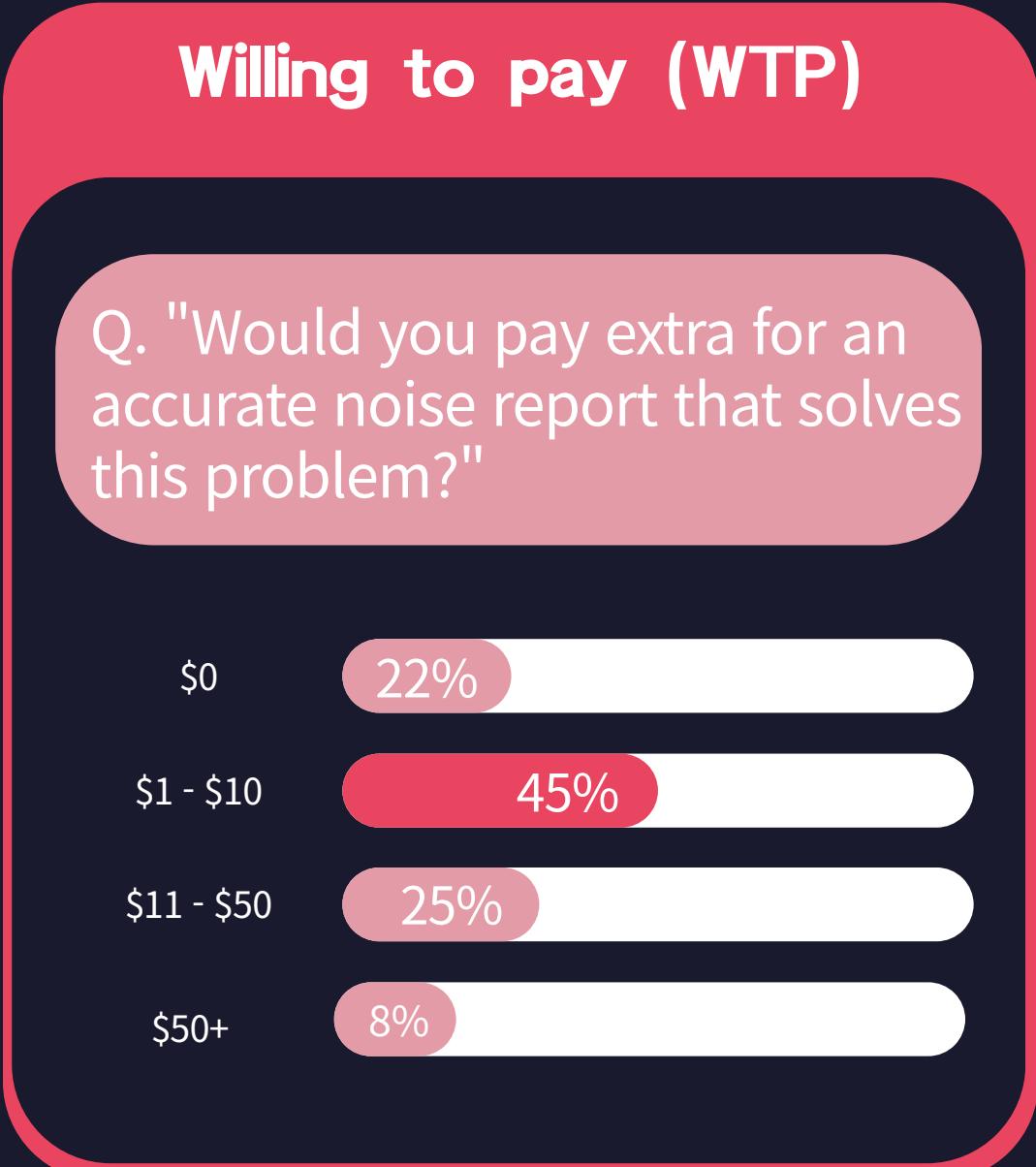
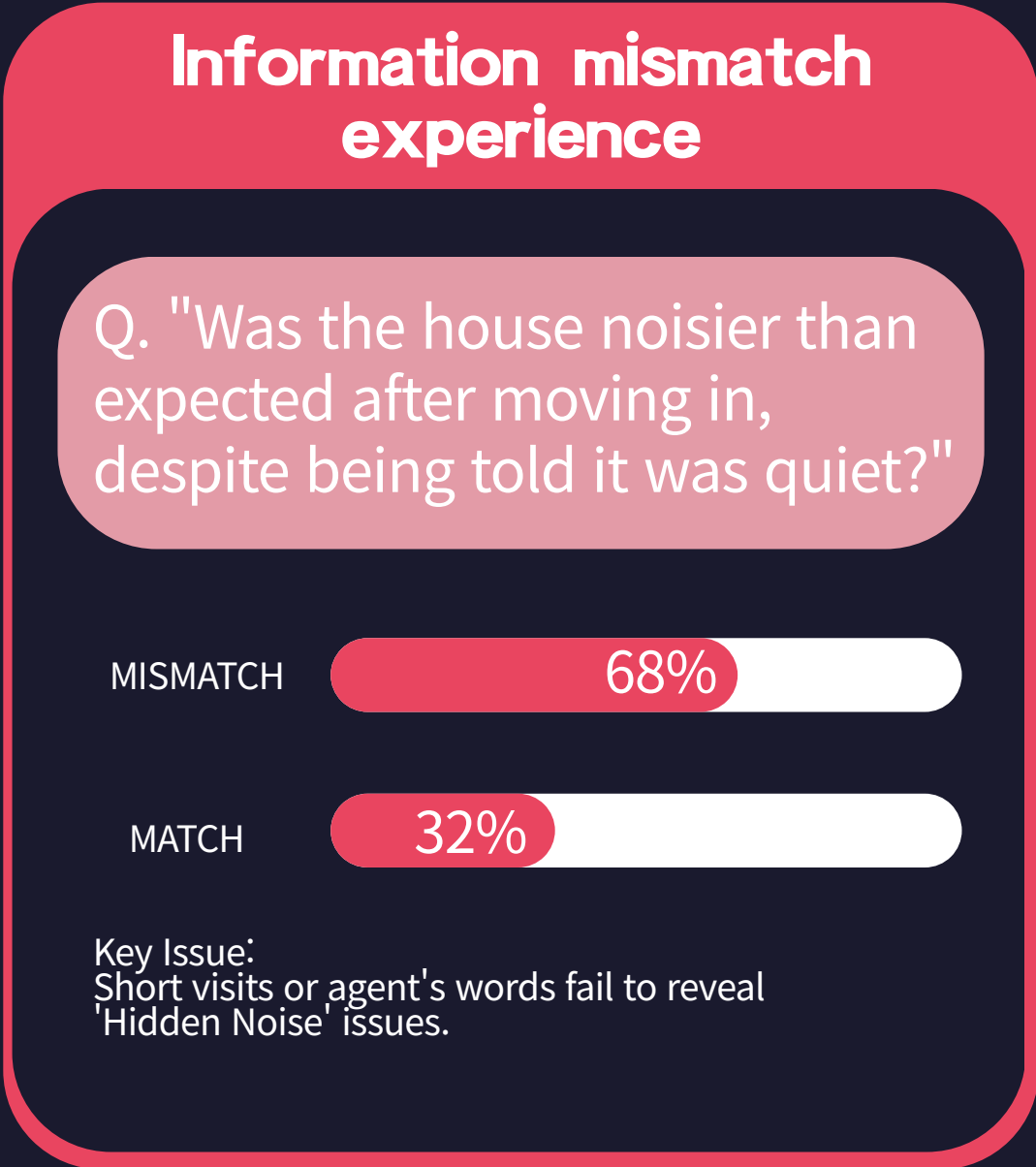
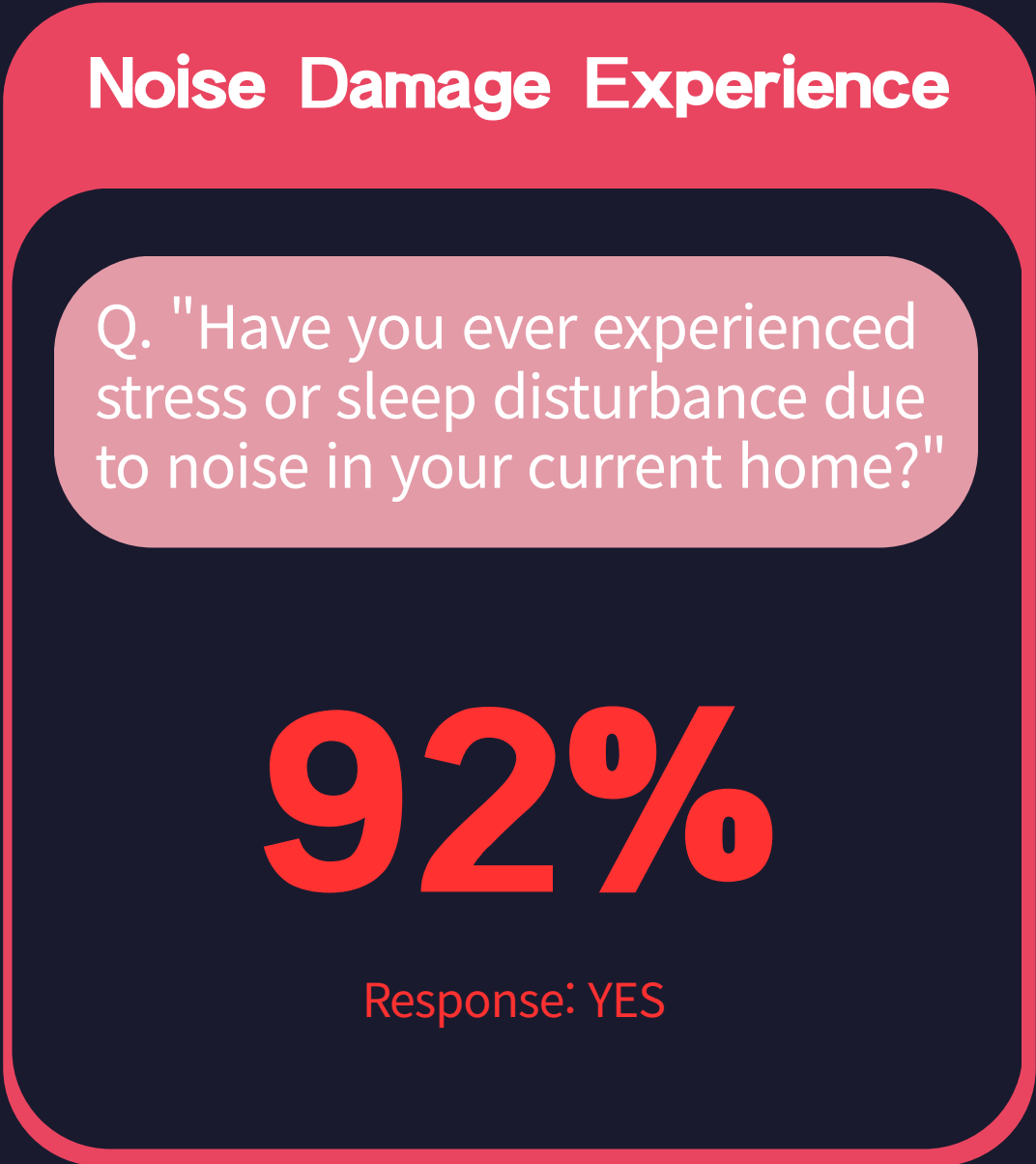
- Sound Scout uses our AWS stack backed by a trained AI model to classify and measure sounds in the surrounding area.
- It then stores the information every 30 minutes and provides a map for that timescale allowing users to check the data later in the day.



Validation Plan_Quantifying the Problem & Value

This is an expected outcome, Through the following questions, we would like to confirm the problem and value.

"Customers will pay want noise information."



A. Pain Experience: 92%_Source : [WHO Environmental Noise Guidelines for the European Region \(2018\)](#)
B. Information Mismatch: 68%_Source : [George Akerlof \(1970\) - "The Market for Lemons" \(Nobel Prize in Economics\)](#)
C. Willingness to Pay (WTP): 78%_Source : [George Akerlof \(1970\) - "The Market for Lemons" \(Nobel Prize in Economics\)](#)